

Artificial Intelligence - 2

ASSIGNMENT D6-V1

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Scenario & Modelling Approach

A free-climbing robot inspects external ISS hardware.

- Traverse a graph of handrail locations
- At each stop: inspect the component present, or move on
- Match the correct sensor to the component's modality
- Manage two coupled resources: battery & data storage
- Return to dock, upload buffered data before the mission ends

TWO FORMULATIONS, ONE SCENARIO

Q1-Classical PDDL

:typing, :strips, :negative-preconditions, :numeric-fluents
discrete, instantaneous actions only

Q2-PDDL+

adds :time, :continuous-effects, :disjunctive-preconditions
durative activities, continuous processes, autonomous events, full
orbital day/night cycle

Same mission logic underneath-Q2 re-expresses *when* and *how fast* things happen, not *what* needs to happen.

robot
location → docking
components
sensors

Distinct first-class types $\Rightarrow$
grounder rejects invalid substitutions
before search,
keeping the grounded action set small.

- **Location:** robot_at, component_at, location_reachable
- **Sensing:** requires_sensor / has_sensor
- **Bookkeeping:** checked_component, data_stored
- **Interlock:** forces an explicit inspect-or-skip decision before the robot can move on
- **Sun:** gates both movement/inspection cost *and* charging rate
- **Numeric budgets:** battery_level, storage_used, data_size

Q1-Classical PDDL

Q1: KEY MODELLING DECISIONS

- **Interlock:** every `move_*` sets `needs_inspection`; only `inspect_*` or `skip_inspection` clears it
- **Sun-aware actions:** `move_sun/move_without_sun`, `inspect_sun/inspect_without_sun`
sun-assisted variants cost only 60% of battery
- **Sensor gate:** wrong hardware onboard $\Rightarrow$ part of the mission silently becomes unsolvable

Q1: RESOURCES & UPLOAD

- **Storage as bin-packing:** `data_size`, `storage_used`, `storage` planner must fit payloads together across a single trip
- **Threshold charging:** `charge_start` only legal once battery ≤ 5 *and* robot is sunlit; charge adds +5 per call, capped at 90
- **Upload = reset:** `upload_data` (dock only) zeroes `storage_used` forces at least one return trip on multi-component missions

Q1: SIMPLE PLAN

One sunlit edge, one component, ample battery (50) & storage (20).

```
0: (move_sun rob1 dock site_a)
1: (inspect_sun rob1 site_a panel_1 camera)
2: (move_sun rob1 site_a dock)
3: (upload_data rob1 dock)
```

Confirms the baseline move → inspect → move → upload loop, and that sun-assisted actions are preferred when available.

Q1: HARD INSTANCE-SETUP

Star topology through a hub; 3 components; radiator_site permanently shaded.

Storage: 14

cap < total payload (23)

Battery: 19

too low to finish without
recharging

Even the best pairing (panel + radiator = 13) still doesn't fit everything-a valid satisficing plan need not be storage-optimal.

Q1: HARD PLAN-THREE ROUND TRIPS

57-step plan, one dock round-trip per component:

1. **Scout & charge:** probe antenna branch (no inspection), then a full charge cycle at hub
2. **Shaded radiator branch:** dive in on banked charge, inspect with `thermal_camera`
3. **Upload 1** at dock, partial recharge
4. **Antenna branch** (2nd visit): inspect for real, **upload 2**
5. **Panel branch:** inspect, top-up charge, **upload 3**-goal reached

Q1: THE “WASTEFUL” DETOUR, EXPLAINED

No wait action exists-battery only drops via movement or inspection.

Step	Move	Battery
start	-	19
0	dock $\rightarrow$ hub (sun)	$19 - 6 = 13$
2	hub $\rightarrow$ antenna_site (sun)	$13 - 6 = 7$
4	antenna_site $\rightarrow$ hub (sun)	$7 - 6 = 1$

The antenna round-trip inspects *nothing*-its only purpose is burning charge down to the ≤ 5 threshold while landing back on a *sunlit* node, so `charge_start` becomes legal.

Q1: WHY NOT GO STRAIGHT TO THE RADIATOR?

Direct route dock $\rightarrow$ hub $\rightarrow$ radiator_site, no charging stop:

■ Battery 19 $\xrightarrow{\text{sun}, -6}$ 13 $\xrightarrow{\text{no sun}, -8}$ 5

Stranded at battery 5:

- Can't inspect-needs battery ≥ 10
- Can't leave-move back costs 8
- Can't charge-radiator_site has no sun

$\Rightarrow$ the detour through hub isn't wasteful-it's the *only* way to survive the shaded leg.

Q2-PDDL+

Q2: DURATIVE ACTIVITIES

Each Q1 action becomes a start action + autonomous end event:

- `move_start / move_end`
- `inspect_start / inspect_end`

Start sets busy and resets a timer; a `:process` advances that timer; the matching event fires once the required duration is reached-the canonical PDDL+ `start/process/end` idiom.

Q2: CONTINUOUS PROCESSES

- `move_clock / inspect_clock`-timers gating the `_end` events
- `drain_battery`-during transit
- `collect_data`-battery & storage cost spread over inspection duration
- `charge`-always-on, scaled by `sun_exposure` & `solar_factor`
- `orbit_tick`-the master clock
- `transmit_data`-during upload, gated by the comm window

Q2: AUTONOMOUS EVENTS

- Activity closure: `move_end`, `inspect_end`, `upload_complete`
- `dead-battery` ≤ 0 , disables the robot (prunable by the heuristic)
- Day/night cycle-45 min / 45 min, orbit index wraps every 16 orbits
- Solar factor-9 events stepping $\{0, 0.2, \dots, 1.0\}$ as $P \propto \cos \theta$
- Sun exposure-5 events recompute $\{0, 1, 2\}$ (shade / transit / full sun)
- Comm window-opens 10 min per orbit, first 3 orbits of each day

Q2: A DELIBERATE ASYMMETRY

Why is battery exhaustion an **event** but a full buffer just a **precondition**?

- **Battery** can hit zero *unprompted*, mid-activity-must interrupt whatever is happening $\Rightarrow$ modelled as the dead event
- **Storage** only blocks *starting* a new inspection-collect_data's rate is only ever applied to an inspection already checked to fit $\Rightarrow$ modelled as a precondition on `inspect_start`

Q2 PROBLEM INSTANCES

p1_simple: dock, loc_a, loc_b mutually reachable, fully lit
generous battery/storage-only real constraint is comm-window *timing*

p2_hard: forced relay (loc_b only via loc_a), loc_b permanently shaded
low starting battery (18/30)-must banked-charge before the shaded leg

Q2: SIMPLE PLAN

```
00:00-24:00    waiting (align to first comm window)
24:00-24:08    transit dock -> loc_a
24:08-24:15    inspect panel1 @ loc_a (camera)
24:15-24:17    transit loc_a -> dock
24:17-25:24    wait for transmit window, then upload
25:24-25:41    transit dock -> loc_b
25:41-25:44    inspect panel2 @ loc_b (spectrometer)
25:44-25:46    transit loc_b -> dock, final upload completes
```

≈107s of real activity; the dominant chunk (≈67s) is simply *waiting* at dock for the comm window-battery/storage never bind here.

Q2: HARD PLAN HIGHLIGHTS

- loc_a visited **three** times; panel1 inspection deferred twice
- Robot pre-charges at sunlit loc_a *before* committing to the shaded leg-no top-up possible once at loc_b
- Dock arrivals must also line up with an open comm window

4528s total, $\approx$ 31s real activity

under 1%-over 99% is charge/comm-window waiting

Comparison & Conclusion

Q1 vs. Q2

	Q1 (classical)	Q2 (PDDL+)
Duration	instantaneous, lump cost	start/process/end, continuous
Battery	discrete decrease per action	continuous drain/collect/transmit
Charging	discrete action, threshold ≤ 5	always-on process, sun-scaled
Sun/shadow	static boolean	dynamic exposure + orbital angle
Day/night & comms	not modelled	full orbit, day/night, comm windows
Battery-critical	implicit (inapplicable actions)	explicit dead event
Feasibility driver	route/order vs. storage & battery	<i>plus</i> exact orbital/comm timing
Idle time	none	dominant (>90% of both plans)

CONCLUSION

- Q1 validates the core mission logic: routing, sensor-matched inspection, bounded storage, threshold charging
- Q2 preserves that logic while adding real-valued durations, continuous dynamics, and an orbital day/night/comm cycle
- Q1's hard plan is shaped by **resource bin-packing**; Q2's is shaped almost entirely by **orbital/comm-window timing**
- PDDL+ is strictly more expressive, but at a much larger search cost
- A validated foundation to extend toward a full maintenance-planning domain

Thank you

Backup

BACKUP: FULL Q1 HARD PLAN I

```
0: (move_sun rob1 dock hub)
1: (skip_inspection rob1)
2: (move_sun rob1 hub antenna_site)
3: (skip_inspection rob1)
4: (move_sun rob1 antenna_site hub)
5: (charge_start rob1 hub)
6-12: (charge rob1 hub) x7
13: (stop_charging rob1)
14: (skip_inspection rob1)
15: (move_without_sun rob1 hub radiator_site)
16: (inspect_without_sun rob1 radiator_site radiator_1
    thermal_camera)
17: (move_without_sun rob1 radiator_site hub)
18: (skip_inspection rob1)
19: (move_sun rob1 hub dock)
20: (upload_data rob1 dock)
21: (charge_start rob1 dock)
22: (charge rob1 dock)
23: (stop_charging rob1)
24: (skip_inspection rob1)
25: (move_sun rob1 dock hub)
26: (charge_start rob1 hub)
```

BACKUP: FULL Q1 HARD PLAN II

```
27-30: (charge rob1 hub) x4
31: (stop_charging rob1)
32: (skip_inspection rob1)
33: (move_sun rob1 hub antenna_site)
34: (inspect_sun rob1 antenna_site antenna_1 highres_camera)
35: (move_sun rob1 antenna_site hub)
36: (skip_inspection rob1)
37: (move_sun rob1 hub dock)
38: (upload_data rob1 dock)
39: (charge_start rob1 dock)
40-43: (charge rob1 dock) x4
44: (stop_charging rob1)
45: (skip_inspection rob1)
46: (move_sun rob1 dock hub)
47: (skip_inspection rob1)
48: (move_sun rob1 hub panel_site)
49: (inspect_sun rob1 panel_site panel_1 camera)
50: (move_sun rob1 panel_site hub)
51: (skip_inspection rob1)
52: (charge_start rob1 hub)
53: (charge rob1 hub)
54: (stop_charging rob1)
55: (move_sun rob1 hub dock)
```

BACKUP: FULL Q1 HARD PLAN III

```
56: (upload_data rob1 dock)
```

BACKUP: Q2 HARD PLAN-MILESTONE TIMELINE

1. Wait $\sim 306\text{s}$ $\rightarrow$ transit to `loc_a` $\rightarrow$ skip `panel1` $\rightarrow$ wait $\sim 1327\text{s}$ (pre-charge)
2. Transit to shaded `loc_b` $\rightarrow$ wait $\sim 1435\text{s}$ $\rightarrow$ inspect `panel2`
3. Return via `loc_a` (skip `panel1` again) to dock $\rightarrow$ upload 1 + recharge ($\sim 59\text{s}$)
4. Transit to `loc_a` $\rightarrow$ wait $\sim 1369\text{s}$ $\rightarrow$ inspect `panel1` $\rightarrow$ return to dock, final upload

Total: 4528s; real activity: $\sim 31\text{s}$.